

Unit 2 :-Chapter 1 Quantity Theory of Money

(Fisher and Cambridge Approaches)

1. Introduction

Quantity Theory of Money explains the relationship between quantity of money and the general price level. It states that changes in money supply influence the value of money and prices.

2. Fisher's Quantity Theory of Money (Transaction Approach)

Irving Fisher, an American economist, gave the mathematical form of the Quantity Theory of Money in his famous 1911 book "The Purchasing Power of Money". Fisher explained the relationship between the quantity of money and the general price level. According to Fisher, money is mainly used as a medium of exchange, and the value of goods and services exchanged in an economy must be equal to the total money spent on those transactions. Fisher's Quantity Theory of Money (Transaction Approach)

Fisher's Equation of Exchange:

$$MV = PT$$

Where,

M = Money Supply

It refers to the total quantity of money available in the economy for carrying out transactions.

V = Velocity of Circulation of Money

It refers to the average number of times a unit of money changes hands during a given period.

P = General Price Level

It represents the average price of goods and services in the economy.

T = Total Volume of Transactions

It represents the total number or value of goods and services exchanged during a given period.

Meaning of Fisher's Equation:

The equation states that the total money expenditure in an economy (MV) must be equal to the total money value of all transactions (PT).

Therefore:

Total Money Expenditure = Total Value of Transactions

$$MV = PT$$

The left side (MV) shows the total amount of money spent in the economy, while the right side (PT) shows the total value of goods and services purchased.

Determination of Price Level:

From Fisher's equation:

$$P = MV / T$$

The price level depends on:

1. Quantity of money (M)
2. Velocity of circulation of money (V)
3. Volume of transactions (T)

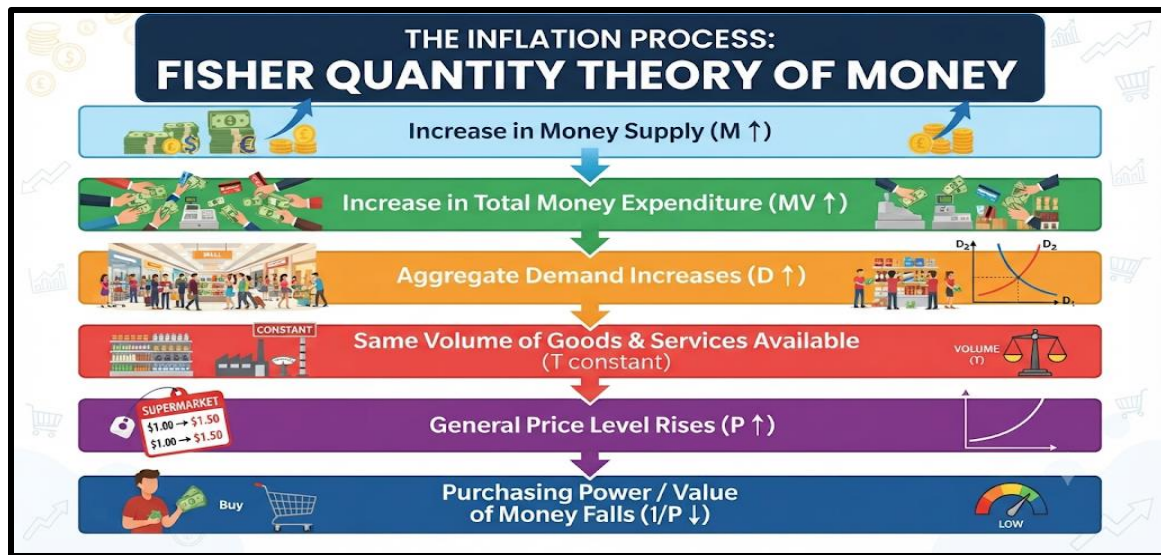
Assuming V and T remain constant:

$$P \propto M$$

This means that the price level changes in the same proportion as the quantity of money.

- **Relationship Between Money Supply and Price Level:**

According to Fisher, an increase in money supply increases total expenditure in the economy. Since the volume of goods and services remains unchanged, the extra purchasing power increases demand and results in a rise in the general price level.



Effect of Doubling of Money Supply:

Suppose:

Initial condition:

$$M = ₹1000$$

$$V = 5$$

$$T = 500$$

Price Level:

$$P = MV/T$$

$$P = (1000 \times 5) / 500$$

$$P = 10$$

Now, if money supply doubles:

$$M = ₹2000$$

Keeping V and T constant:

$$P = (2000 \times 5) / 500$$

$$P = 20$$

Therefore:

Money Supply:

₹1000 → ₹2000 = 100% increase

Price Level:

10 → 20 = 100% increase

Hence, according to Fisher's theory:

If money supply doubles, the price level also doubles, provided velocity of money and volume of transactions remain constant.

Relationship Between Money Supply and Value of Money:

The value of money is the purchasing power of money and is measured as the reciprocal of the price level.

Value of Money = $1/P$

When the price level increases, the value of money decreases.

Therefore:

Price Level ↑ → Value of Money ↓ (inverse relationship) (Price level increases-Value of money decreases)

Example:

Before increase in money supply:

$P = 10$

Value of Money = $1/10$

After doubling of money supply:

$P = 20$

Value of Money = $1/20$

Thus, when money supply doubles:

- Price level becomes double.
- Purchasing power of money becomes half.

The complete relationship is:

$$M \uparrow \rightarrow P \uparrow \rightarrow 1/P \downarrow$$

More Money $\rightarrow$ Higher Prices $\rightarrow$ Lower Value of Money

Fisher's Extended Equation

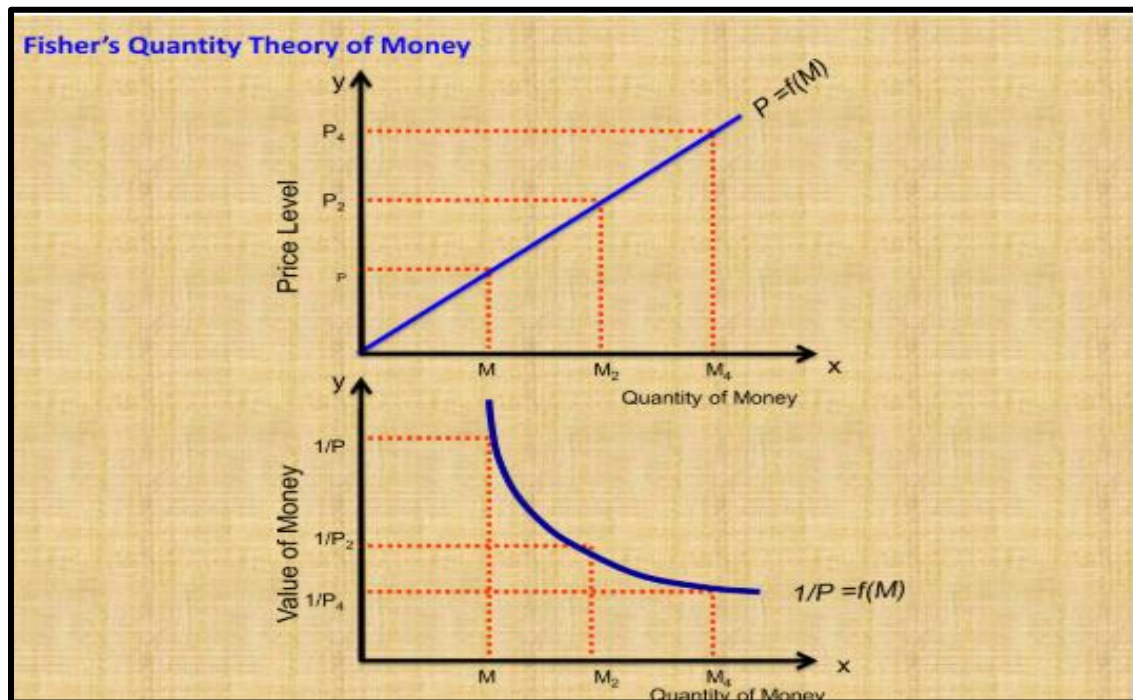
$$MV + M_1V_1 = PT$$

M_1 represents bank deposits/credit money and V_1 represents velocity of bank money.

Fisher's Assumptions

- Velocity of money remains constant.
- Volume of transactions remains constant.
- Money supply changes independently.
- Other factors remain unchanged.
- Money functions mainly as a medium of exchange.

Fisher Diagrams



2. Cambridge Cash Balance Approach

The Cambridge approach focuses on the **demand for holding money balances**. According to Cambridge economists, people do not hold money only for transactions but also because they prefer to keep a part of their income in liquid form. Money is considered both a **medium of exchange and a store of value**.

Cambridge Equation:

$$M_d = kPY$$

Where:

M_d = Demand for money balances

k = Proportion of income people prefer to hold as cash

P = Price level

Y = Real output/national income

The equation shows that the demand for money depends on the **level of income and people's preference for holding cash balances**.

Keynes' Concept of Demand for Money (Liquidity Preference Theory)

Keynes further developed the idea of demand for money by explaining that people hold money for three main motives:

1. Transaction Motive (**M_t**)

- People hold money to meet their **daily buying and selling needs**.
- It depends mainly on **income and the volume of transactions**.
- Higher income leads to a higher demand for transaction balances.

Example: Money kept for purchasing food, paying bills, and regular expenses.

2. Precautionary Motive (**M_p**)

- People hold money to meet **unexpected future needs or emergencies**.
- It acts as a safety reserve.
- Demand for precautionary balances increases with higher income and uncertainty.

Example: Keeping cash for medical emergencies or sudden expenses.

3. Speculative Motive (**M_s**)

- People hold money to take advantage of future changes in **interest rates and bond prices**.

- When interest rates are expected to rise, people prefer holding cash rather than bonds.
- When interest rates are low, people may hold more money expecting better investment opportunities later.

Example: Keeping cash to buy securities when prices become favourable.

Keynesian Demand for Money Equation:

$$M_d = M_{dt} + M_{dp} + M_{ds}$$

Where:

M_{dt} = Transaction demand for money

M_{dp} = Precautionary demand for money

M_{ds} = Speculative demand for money

Features of Cambridge Approach

- Emphasizes **money-holding behaviour** of individuals.
- Gives importance to the **demand for money balances**.
- Uses the **cash balance ratio (k)**.
- Considers money as both a **medium of exchange and a store of value**.
- Recognizes that people hold money based on their **income level and preferences**.
- Provides the foundation for Keynes' **liquidity preference theory**.
- Explains that money demand arises from **transaction, precautionary and speculative motives**.

Keynes' Demand for Money Function

According to Keynes, the demand for money depends mainly on **income (Y)** and the **rate of interest (r)**.

Keynesian Money Demand Function:

$$M_d = f(Y, r)$$

Where:

M_d = Demand for money

Y = Level of income

r = Rate of interest

Explanation:

- **Income (Y):**
Demand for money increases when income rises because people need more money for transactions and precautionary purposes.

- **Rate of Interest (r):**

Demand for money is inversely related to the rate of interest. When the interest rate is high, people prefer investing their money in bonds and other assets, so they hold less cash. When the interest rate is low, people prefer holding more cash.

Relationship:

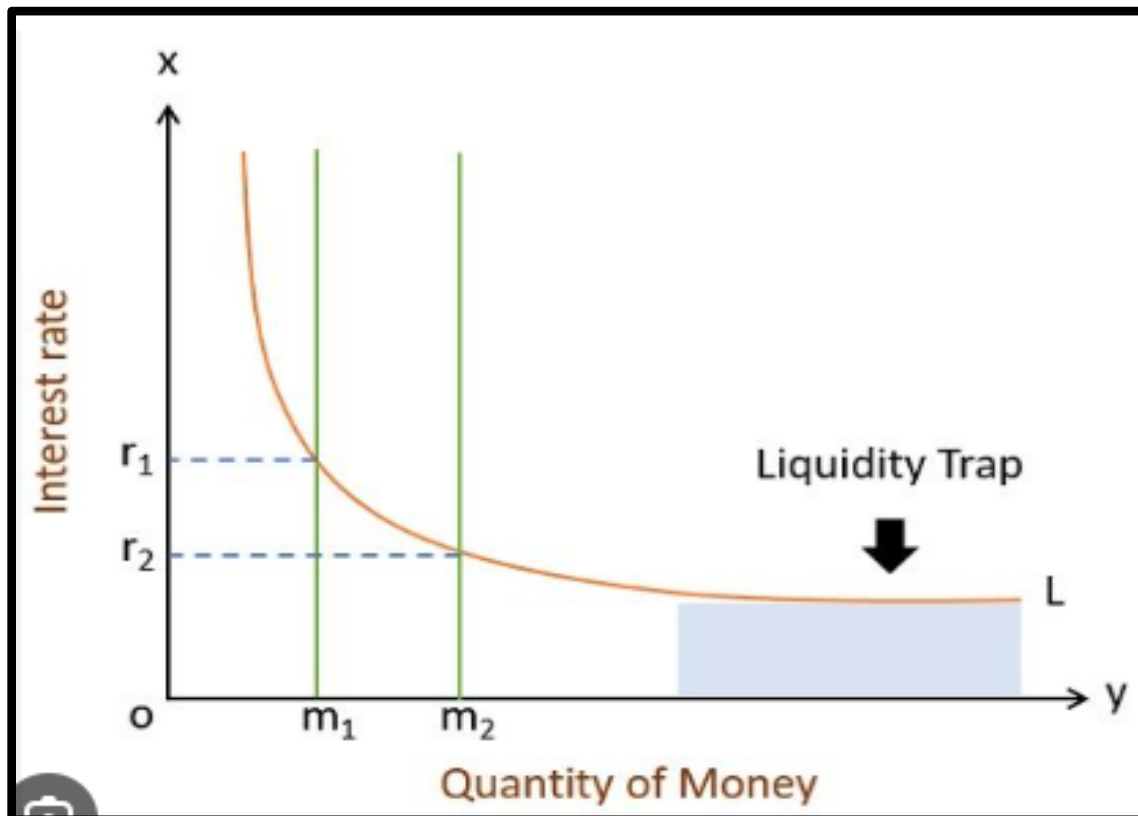
- **Md increases with an increase in Y** → Positive relationship
- **Md decreases with an increase in r** → Negative relationship

Therefore, Keynes expressed money demand as:

$$Md = f(Y, r)$$

This concept expanded the Cambridge approach by explaining that money demand is influenced not only by income and cash balance preference (**k**) but also by **interest rates** and **people's liquidity preference**.

Figure :- Cash holding or Liquidity Preference



Difference between Fisher's Approach and Cambridge Approach

Basis	Fisher's Transaction Approach	Cambridge Cash-Balance Approach
Main focus	Focuses on money used for transactions	Focuses on money people want to hold as cash
Role of money	Money is mainly a medium of exchange	Money is both a medium of exchange and store of value
Equation	$MV = PT$ (or $MV + M'V' = PT$)	$M = kPY$ (Marshall/Pigou version)
Main economists	Irving Fisher	Marshall, Pigou, Robertson and other Cambridge economists
Key variable	V = Velocity of circulation of money	k = Proportion of income held as cash
Approach	Transaction or expenditure approach	Cash-balance or demand-for-money approach

Basis	Fisher's Transaction Approach	Cambridge Cash-Balance Approach
Viewpoint	Looks at the supply and spending of money	Looks at the demand for money balances
Nature of money	Emphasizes how frequently money changes hands	Emphasizes how much money people prefer to keep
Income consideration	Uses total volume of transactions (T)	Uses national income or output (Y)
Velocity	Velocity of money (V) plays an important role	Velocity is not directly used; instead, k ($1/V$) is used
Cash holding	Gives little importance to holding idle cash	Gives special importance to holding cash balances
Human behaviour	Gives less attention to people's desire to hold money	Gives greater importance to people's preference for holding money
Price determination	Price level depends mainly on money	Price level depends on money supply,

Basis	Fisher's Transaction Approach	Cambridge Cash-Balance Approach
	supply, velocity and transactions	income/output and cash-holding preference
Nature of analysis	More mechanical in nature	More realistic and behavioural in nature
Relationship	From $P = \frac{MV}{T}$, where V is the inverse of k ($V = \frac{1}{k}$)	From $P = \frac{M}{kY}$, where k is the inverse of V ($k = \frac{1}{V}$)
Effect of rise in money supply	If V and T remain constant, an increase in M causes a proportionate rise in P	Effect on P also depends on changes in k and real income/output (Y)